
**STRATEGIC INTELLIGENCE FORESIGHT ON RANSOMWARE THREATS
IN STATE-OWNED BANKINGS IN INDONESIA**

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ABSTRACT

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The advancement of artificial intelligence (AI) technology in recent years has been aggressively exploited in cyberattacks, particularly through increasingly sophisticated and massive ransomware deployments. Globally, ransomware incidents have risen by 68% over the past two years (Suvorova, 2023), while Indonesia recorded over one million ransomware activities in 2023 alone, with the financial sector being one of the primary targets (BSSN, 2023). State-owned banks (Bank BUMN) are particularly vulnerable due to their classification as critical information infrastructure and their role in managing the nation's strategic financial assets (Crisanto, Prenio, & Restoy, 2023). This study aims to identify the escalation patterns of AI-powered ransomware attacks against state-owned banks in Indonesia, analyze their vulnerability profiles, and formulate early prevention strategies using a strategic intelligence foresight approach for the period 2025–2029. A qualitative method is applied using the scientific inquiry for intelligence analysis framework (Prunkun, 2010) and a five-phase foresight methodology: event identification, trend analysis, driver mapping, scenario development, and policy roadmap formulation (Saritas, 2016). The 2023 LockBit 3.0 ransomware attack on Bank Syariah Indonesia (Chakravarti, 2023) underscores the urgency of integrating foresight into national banking cybersecurity policy formulation.



Introduction

In the past five years, ransomware attacks in Indonesia's financial sector have escalated significantly. According to the National Cyber and Crypto Agency (BSSN), more than one million ransomware activities were recorded throughout 2023, with the financial sector as a primary target (BSSN, 2023). A critical incident occurred in May 2023 when Bank Syariah Indonesia (BSI) was hit by a LockBit 3.0 ransomware attack, leading to 1.5 terabytes of leaked data and disruptions to digital services such as ATMs and mobile banking (Chakravarti, 2023). In 2024, a similar attack allegedly targeted Bank Rakyat Indonesia (BRI), further emphasizing state-owned banks (Bank BUMN) as recurring strategic targets for cybercriminals (Kompas, 2024).

Bank BUMNs are classified as critical information infrastructure due to their strategic function in managing public funds and serving as the backbone of national payment systems (Crisanto, Prenio, & Restoy, 2023). Disruptions to their digital infrastructure can pose fiscal instability and erode public trust in national financial systems. Given their wide operational scope and transaction volume, Bank BUMNs have become a prime target for economically motivated cyber threats.

As noted by Simorangkir, Purba, and Hartati (2025), Indonesia's banking cybersecurity systems continue to struggle against advanced threats like zero-day exploits and AI-powered ransomware. This situation calls for a shift from reactive, technical responses to anticipatory and strategic approaches. Hence, the integration of strategic intelligence and foresight frameworks becomes essential in crafting cybersecurity mitigation policies that consider both past incidents and future threat dynamics.

This study seeks to formulate mid-term cybersecurity strategies to counter the rising threat of AI-enhanced ransomware targeting Indonesian state-owned banks between 2025 and 2029. Utilizing strategic foresight as a core methodology, this paper offers a proactive framework for policy formulation that is systematic, context-driven, and forward-looking.

Research Methods

This study uses a qualitative method with an exploratory-descriptive approach aimed at understanding the escalation of AI-based ransomware threats to state-owned banks (Bank BUMN) in Indonesia and formulating a strategic response. The main analytical framework employed is the *Scientific Inquiry for Intelligence Analysis* developed by Prunckun (2010), which is designed to structure intelligence-based research with a focus on systematic investigation, validity of sources, and operational relevance.

To complement the qualitative approach, this research also utilizes a foresight method consisting of five stages: event identification, trend analysis, driver mapping, scenario development, and policy roadmap formulation, as suggested by Saritas (2016). These foresight stages are applied to map cybersecurity challenges from 2020 to 2024 and project strategic responses for the 2025–2029 period.

Data collection was conducted through literature analysis of peer-reviewed journals, government regulations, official institutional reports (e.g., BSSN, OJK), and global cybersecurity assessments (e.g., ENISA, CrowdStrike). Additionally, key data points were validated through expert interviews with cybersecurity practitioners, intelligence analysts, and digital security consultants from both state and private sectors.

The data analysis technique employed thematic coding with NVivo to identify patterns of ransomware escalation, institutional vulnerabilities, and gaps in policy response. Each theme is then positioned within the foresight framework to generate policy alternatives based on different future scenarios of cyber threats.

Results and Discussion

Cybersecurity Regulations for the Indonesian Banking Sector

In an effort to strengthen the national financial sector's resilience against digital threats, several regulatory frameworks have been issued by the Financial Services Authority (OJK), the National Cyber and Crypto Agency (BSSN), and the central government. Chief among these is POJK No. 11/POJK.03/2022, which governs the management of information technology in commercial banks. This regulation emphasizes IT governance, risk management, and the implementation of the confidentiality, integrity, and availability (CIA) principles. It also mandates that banks report cybersecurity incidents and conduct regular technical audits (OJK, 2022). Prior to this, POJK No. 38/POJK.03/2016 had addressed risk management in IT usage, although its scope was limited and insufficiently responsive to the growing complexity of digital threats.

In parallel, Presidential Regulation No. 82 of 2022 designates the banking sector as part of the nation's critical information infrastructure, thereby requiring maximum protection. However, its implementation remains general in nature and lacks detailed technical operational guidelines. Additionally, SEOJK No. 29/SEOJK.03/2022 provides technical guidance on IT security, including mandatory vulnerability assessments and penetration testing as systematic preventive measures (OJK, 2022). Another relevant instrument is BSSN Regulation No. 4 of 2021, which outlines a comprehensive cybersecurity risk management framework for electronic system providers, including asset identification, risk analysis, and incident response management (BSSN, 2021).

Despite the existence of these regulatory instruments, they fall short of addressing the latest wave of sophisticated cyberattacks—particularly those that exploit artificial intelligence (AI) to accelerate attacks, bypass security systems, and exploit vulnerabilities in cloud-based infrastructures (CrowdStrike, 2025). For instance, POJK No. 11/POJK.03/2022 does not explicitly regulate AI-driven threats or promote inter-agency collaborative detection mechanisms. Furthermore, the regulation stipulates a three-working-day deadline for incident reporting, which is inadequate when compared to the speed and impact of AI-based attacks that can cause systemic disruptions within hours (National Institute of Standards and Technology, 2024). Real-time threat intelligence sharing mechanisms are also not yet operationalized, even though such systems are vital for containing widespread, coordinated attacks (CONCORDIA, 2021).

Substantively, these regulations reflect the regulators' commitment to developing a solid cybersecurity governance structure for the banking sector. However, at the implementation level, challenges remain due to weak inter-agency coordination, delayed incident reporting, and the lack of medium-term policy designs based on scenario forecasting and risk projections. The current approach remains largely reactive and has not kept pace with the evolving nature of ransomware threats, which are increasingly adaptive, multilayered, and rapidly evolving due to support from intelligent technologies (Cybersecurity and Infrastructure Security Agency, 2024).

Ransomware Threats to the Banking Sector

As Indonesia's digital ecosystem continues to evolve in complexity, state-owned banks (Bank BUMN) are increasingly exposed to escalating ransomware threats augmented by artificial intelligence (AI). What were once static attack patterns have now shifted to become adaptive and dynamic, making early detection significantly more difficult. Advanced ransomware variants such as RansomAI have been reported to employ reinforcement learning techniques, enabling the malware to autonomously adjust its propagation strategies based on the target network's structure. This facilitates highly efficient infiltration with unpredictable attack vectors (Zhou, Lin, & Patel, 2023). Automated lateral movement and asset mapping further allow threat actors to exploit system vulnerabilities within minutes, thereby diminishing the effectiveness of traditional manual response mechanisms commonly relied upon by banks (CrowdStrike, 2025).

According to the *Indonesia Cybersecurity Landscape Report 2024*, a total of 514,508 ransomware activities were detected in national cyberspace during that year. The LockBit variant accounted for the highest number of incidents, with 102,798 cases. The financial sector, including banking, was among the most vulnerable targets (Badan Siber dan Sandi Negara, 2024). Attack techniques ranged from phishing and administrator privilege abuse to exploiting unpatched application vulnerabilities. The threat is especially critical for institutions like Bank BUMN, which manage vast digital assets and hold public mandates for financial operations and national payment systems.

From a financial standpoint, ransomware incidents in Bank BUMN have led to substantial losses. A 2025 report from Halcyon estimates that the average cost of a ransomware incident in the global financial sector reached USD 6.08 million in 2024, covering system recovery, IT infrastructure replacement, ransom payments, and service

disruption losses. A prominent example is the EquiLend case, where operational paralysis forced financial institutions to revise capital reserves and absorb rising operational expenses (Reuters, 2024). In Indonesia, Bank Syariah Indonesia (BSI) experienced a severe ransomware attack in May 2023. The LockBit 3.0 group claimed responsibility, allegedly stealing 1.5 terabytes of customer data and demanding a ransom of IDR 200 billion. The attack disrupted key banking services such as ATMs and mobile banking for several days, affecting operational continuity and customer trust (Kompas.id, 2023). If such incidents were to compromise core banking systems across other Bank BUMNs, the potential operational and financial instability would be even more severe.

Beyond financial losses, reputational damage poses a long-term threat. Data breaches and service outages erode public confidence in the government's ability to safeguard sensitive financial information. This reputational impact not only diminishes customer loyalty but also undermines the institution's appeal to potential investors and its internal financial stability (Mavroeidis & Bromander, 2019). Given the strategic role of Bank BUMNs as instruments of national fiscal policy and development, reputational crises can escalate into systemic risks.

The absence of timely regulatory updates and the slow adoption of predictive technologies exacerbate the threat landscape. Current domestic regulations lack explicit provisions addressing AI-based threats, offer insufficient real-time incident reporting mechanisms, and fail to establish a robust threat intelligence sharing framework between institutions (CONCORDIA, 2021; CIS, 2023). In worst-case scenarios, coordinated ransomware attacks targeting multiple Bank BUMNs simultaneously could trigger fiscal shocks and diminish public trust in digital financial infrastructure (Linkov & Kott, 2019).

Addressing these vulnerabilities requires the adoption of international standards such as ISO/IEC 27035, which provides a structured framework for security incident management, including detection, containment, recovery, and post-incident analysis (International Organization for Standardization, 2018). However, most Indonesian regulations, including POJK and Presidential Regulations, have yet to fully incorporate these standards into institutional technical governance. The lack of early containment protocols and weak regulatory coordination significantly delay recovery efforts and heighten the risk of greater losses.

Therefore, cybersecurity strategies in Indonesia's state-owned banking sector must move beyond technical interventions alone. They must also promote adaptive regulatory transformation in response to evolving AI-driven threat dynamics. Key elements should include scenario-based mitigation planning, enhanced internal cyber intelligence capacity, and the alignment of domestic policies with global cybersecurity standards to build a comprehensive and resilient digital financial architecture.

Foresight Horizon Scanning

Event Identification

Within the horizon scanning framework, the event identification phase aims to map real-world incidents that have had direct implications on the information security landscape of Indonesia's state-owned banking sector. A review of the 2020–2024 period reveals two major incidents that exemplify the rising intensity and sophistication of ransomware attacks targeting state financial institutions.

The first incident occurred in May 2023, when the LockBit 3.0 ransomware successfully breached the digital security infrastructure of Bank Syariah Indonesia (BSI). The attack resulted in the exfiltration of approximately 1.5 terabytes of sensitive data and caused widespread disruptions to digital banking services, including mobile applications, ATM networks, and internal systems, for more than five days (Chakravarti, 2023). The impact was immediately felt by millions of customers who lost access to essential banking services. Given BSI's position as the consolidated entity of Indonesia's national Islamic financial institutions, this breach significantly eroded public confidence in the institution's preparedness to manage high-level cyber risks.

The second event involved a suspected ransomware attack on Bank Rakyat Indonesia (BRI) in late 2024. While official confirmation regarding data leaks was not issued, media reports indicated disruptions across several digital banking services. The bank initiated an internal investigation to assess the cause and scope of the incident. This case further reinforced the notion that AI-powered ransomware threats had begun to compromise systems within BUMN banks, which were previously regarded as resilient against large-scale cyberattacks (Kompas, 2024).

These incidents are further supported by the 2024 Indonesian Cybersecurity Landscape report, which recorded 514,508 ransomware-related activities within national cyberspace during the year. Among them, the LockBit variant remained the most active, with 102,798 incidents detected and traced by national threat monitoring systems (Badan Siber dan Sandi Negara, 2024). This data affirms that ransomware has emerged as one of the fastest-growing and most aggressive forms of cyberattacks, targeting critical sectors such as finance.

The cases involving BSI and BRI not only reflect an increase in attack frequency but also indicate a shift toward more systemic and coordinated attack patterns.

Cybercriminals now exploit internal system vulnerabilities using adaptive infiltration strategies, with AI technologies enabling them to map and penetrate critical pathways within digital infrastructures with unprecedented speed and precision.

Consequently, these two events serve as empirical anchors in the strategic intelligence-based foresight process. Their identification not only characterizes the current threat environment but also provides a foundational basis for constructing mid-term mitigation scenarios for the 2025–2029 period. This ensures a proactive and contextually relevant approach to early-stage cyber risk prevention planning.

Trend Analyst

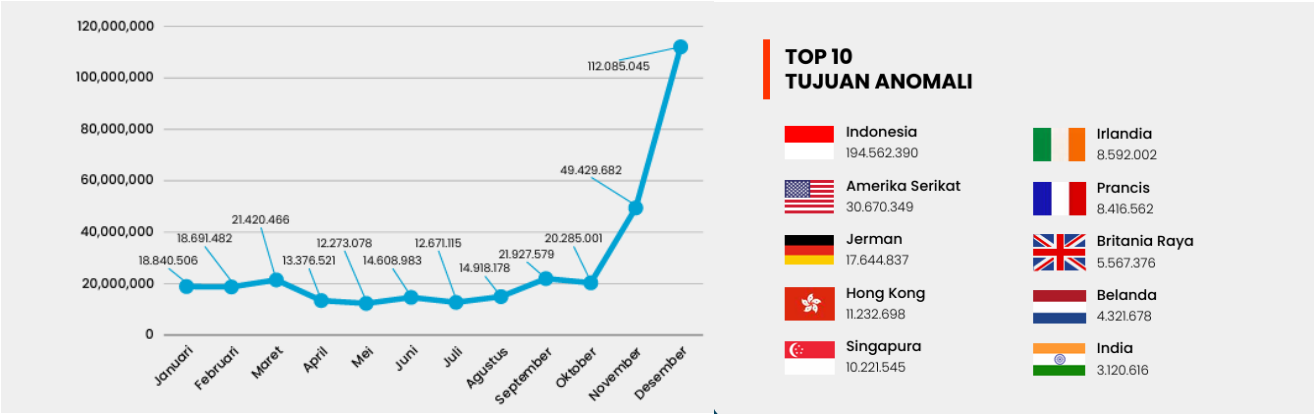


Figure 2.1. Anomalous Cyber Attack Traffic in Indonesia

Source: Indonesian Cybersecurity Landscape Report, 2024

Referring to the *Indonesian Cybersecurity Landscape Report 2024*, the volume of anomalous cyber traffic in Indonesia showed a marked upward trajectory throughout the year, with a particularly sharp increase during the final months. Monthly data from January to October 2024 fluctuated between 12 and 21 million activities. However, there was a significant spike in November, reaching 49,429,682 activities, and a dramatic peak in December with 112,085,045 recorded anomalies. This twofold surge within a single month indicates a critical escalation trend in cyber threat activities (Badan Siber dan Sandi Negara, 2024).

This trend suggests that national digital systems face heightened pressure at the end of the year, likely linked to increased transaction volumes, reduced cybersecurity vigilance during long holiday periods, and the exploitation of unresolved system vulnerabilities. The exponential rise underscores the urgent need to enhance early detection mechanisms—not only to reactively manage incidents but to proactively forecast seasonal threat cycles based on recurring attack patterns.

Beyond its temporal dynamics, the data also highlight Indonesia's global position as the country with the highest recorded cyber anomaly traffic in 2024, amounting to 194,562,390 activities. This figure significantly surpasses those of the United States (36,670,349), Germany (17,644,837), and Singapore (10,221,545). Such dominance signals a substantial exposure of national digital infrastructure to a wide range of attacks, *Jurnal Indonesia Sosial Teknologi*, Vol. x, No. x, Bulan Tahun

including ransomware, affirming Indonesia's status as a primary target within the global threat landscape (Badan Siber dan Sandi Negara, 2024).

Within the foresight framework, these findings strengthen the importance of trend analysis in identifying recurring and progressive threat patterns. The nonlinear escalation observed at year-end offers a strategic signal for mid-term mitigation planning. It also reinforces the need for integrated temporal monitoring mechanisms, particularly within state-owned banks, which handle large volumes of transactions, are closely tied to national financial systems, and exhibit high exposure to systemic cyberattacks.

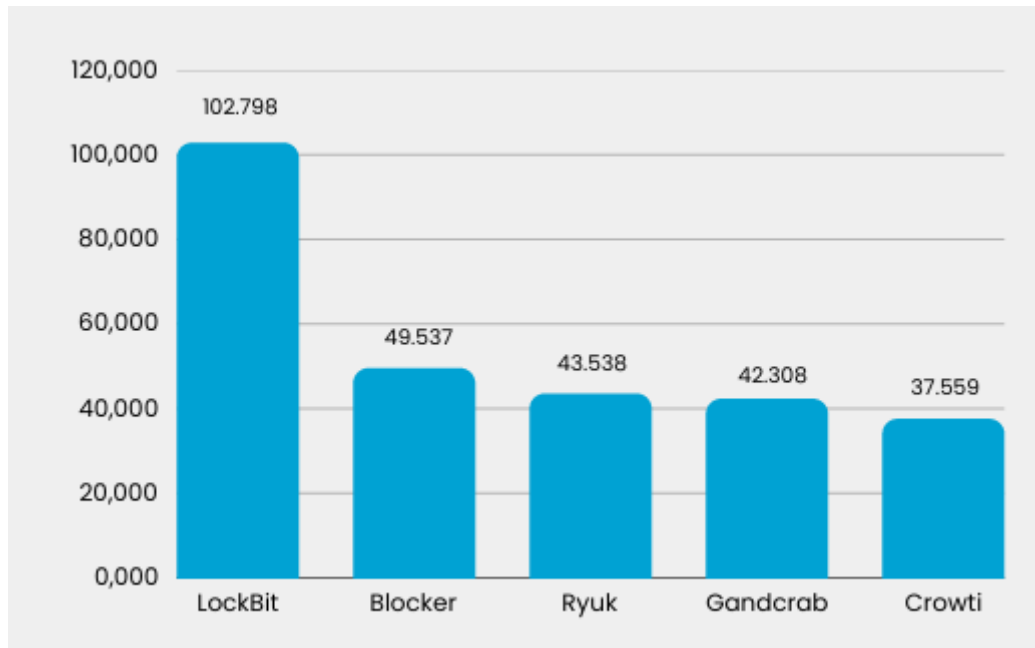


Figure 2.2. Number of Ransomware Activities by Variant in Indonesia in 2024
(Source: National Cyber and Crypto Agency [BSSN], 2024)

According to the *Indonesian Cybersecurity Landscape Report 2024*, a total of 514,508 ransomware-related activities were detected within the national cyberspace throughout the year (Badan Siber dan Sandi Negara, 2024). Ransomware is a type of malicious software designed to encrypt a victim's data or system, followed by a demand for payment in exchange for restoring access. These attacks target various layers of digital users, ranging from individuals to critical institutions, and often cause severe operational, financial, and reputational damage to the affected entities.

A bar chart in the report highlights the five most active ransomware variants in Indonesia during 2024. LockBit ranked highest with 102,798 activities, significantly surpassing other variants. This was followed by Blocker (49,537 activities), Ryuk (43,538), GandCrab (42,308), and Crowti (37,559). LockBit's dominance suggests its superior penetration capability and operational effectiveness in exploiting system vulnerabilities, particularly in complex, high-volume environments such as the financial sector's digital infrastructure.

The concentration of activity around specific variants indicates that ransomware evolution is not solely a matter of quantity but also involves technical consolidation. Certain strains are becoming more refined and harder to counter, reflecting cybercriminals' growing preference for aggressive and high-impact vectors. This pattern is a critical signal within the foresight framework's trend analysis stage, as it reveals threat evolution trajectories and informs strategic detection and response planning.

In the context of state-owned banking institutions (Bank BUMN), the implications are especially significant due to the high volume of data, transaction traffic, and dependency on digital systems. Attacks involving destructive variants such as LockBit and Ryuk could result in widespread service disruption, loss of public trust, and substantial operational costs tied to system recovery and re-securitization efforts. Therefore, there is an urgent need for structured anticipatory policies based on active variant trend analysis, along with the enhancement of targeted detection systems specifically tailored to the threat landscape of the national banking sector.

Driver 1: Automation and Increasing Complexity of AI-Powered Cyberattacks

The rapid advancement of artificial intelligence (AI) has transformed ransomware from a static threat into an adaptive and automated cyber weapon. Contemporary variants are no longer limited to executing data encryption alone; they can now perform real-time scanning to identify system vulnerabilities, move laterally across network environments, and dynamically adjust their infection strategies based on the specific profile of the targeted entity. Notably, strains such as LockBit and Ryuk have integrated machine learning capabilities to optimize attack precision and evade conventional signature-based detection systems (Zhou, Lin, & Patel, 2023).

The *CrowdStrike Threat Report 2025* highlights that over 30% of global ransomware incidents now incorporate AI-driven automation toolkits. These include fileless infection techniques, stealth execution protocols, and auto-deployment payload systems that enable ransomware to spread with minimal human intervention. As a result, the time-to-compromise—the duration required to disable a targeted system—has drastically decreased, from several hours to mere minutes (CrowdStrike, 2025).

This trend underscores the urgency for financial sector cybersecurity policies—especially within state-owned banks—to adapt to the heightened speed and sophistication of modern attacks. Traditional mitigation approaches that lack predictive frameworks will struggle to respond effectively to these rapidly evolving threat vectors. Consequently, future-proof strategies must be designed with agility, real-time detection capabilities, and AI-oriented defense mechanisms to keep pace with the adversary's technological advantage.

Driver 2: Policy Fragmentation and Weaknesses in Cyber Intelligence Sharing

Despite the enactment of regulatory frameworks such as POJK No. 11/POJK.03/2022 and Presidential Regulation No. 82 of 2022 in response to escalating cyber risks, Indonesia continues to face challenges in inter-agency coordination and the absence of an operational national threat intelligence sharing system. The lack of integration among regulatory bodies, industry stakeholders, and cybersecurity service providers has resulted in slower detection and response processes for systemic cyberattacks.

Research has shown that the inability to establish real-time intelligence exchange—particularly concerning metadata, as well as tactics, techniques, and procedures (TTPs)—remains a major obstacle in mitigating ransomware attacks across the financial sector, especially in the Asian context (CONCORDIA, 2021). In practice, this deficiency causes threat responses to operate in silos, rather than reflecting the collective response capacity required to counter coordinated cyber threats.

From a foresight perspective, the fragility of cyber intelligence-sharing infrastructure must be viewed as a critical driver that heightens systemic risks within state-owned banking institutions. If left unaddressed, this gap could severely compromise national cybersecurity resilience. Accordingly, this issue demands the development of a collaborative and cross-sectoral policy framework to enable integrated threat intelligence coordination.

Driver 3: The Industrialization of Cybercrime through Ransomware-as-a-Service (RaaS)

The industrialization of cybercrime, particularly through Ransomware-as-a-Service (RaaS) schemes, has drastically lowered the entry barrier for conducting digital attacks. Threat actors no longer require advanced technical expertise to launch ransomware campaigns; instead, they can purchase or rent ready-made ransomware kits from dark web marketplaces. These kits often come with user-friendly graphical interfaces, detailed instructional manuals, and integrated crypto-based systems for secure transactions and communications.

Recent studies indicate that over 60% of ransomware campaigns targeting the financial sector now originate from RaaS models. Subscription costs for such services typically range between USD 200 and 300 per month, yet they can inflict substantial financial and reputational damage on victim institutions (Kumar & Upadhyay, 2023).

This model significantly intensifies the threat landscape by democratizing access to powerful attack tools, enabling a wider array of adversaries—including those with minimal resources—to conduct high-impact operations. In the context of foresight, the proliferation of RaaS serves as a strategic signal for the need to adopt predictive frameworks capable of simulating a diverse threat spectrum—from state-sponsored groups to opportunistic cybercriminals leveraging commercialized attack platforms.

Scenario Planning

X-Axis – Adaptive Capability of Cybersecurity Systems

This axis represents the capacity of Indonesia's state-owned banking institutions (Bank BUMN) to develop and adjust their cybersecurity defense systems in response to the growing complexity of digital threats—particularly AI-powered ransomware attacks. It encompasses several dimensions, including internal regulatory frameworks, operational preparedness, and the deployment of adaptive detection and incident response mechanisms. A key benchmark of institutional adaptability lies in the maturity of implementing international information security standards such as ISO/IEC 27001 and ISO/IEC 27035, as well as the institution's ability to continuously update its system architecture and security protocols (International Organization for Standardization, 2018).

Other critical elements factored into this dimension include the technical readiness of personnel operating within the Security Operations Center (SOC), the functionality of inter-agency threat intelligence sharing systems, and the consistency of internal audits and periodic vulnerability assessments. Institutions that demonstrate a high degree of adaptability typically adopt zero-trust architecture principles and have implemented real-time incident reporting mechanisms to accelerate their response to cyber anomalies (National Institute of Standards and Technology, 2024; CONCORDIA, 2021).

Y-Axis – Offensive Capabilities of AI-Powered Cyberattacks

This axis evaluates the technical sophistication and complexity of cyberattacks utilizing artificial intelligence technologies, particularly ransomware campaigns targeting the banking sector. Attacks along this spectrum are characterized by high levels of automation, stealth techniques, exploitation of zero-day vulnerabilities, and rapid propagation methods such as lateral movement. Notably, variants like LockBit and Ryuk exemplify how AI is being used to enhance infiltration effectiveness while minimizing the likelihood of detection by conventional security systems (Zhou, Lin, & Patel, 2023).

The complexity of these threats is further amplified by the widespread adoption of Ransomware-as-a-Service (RaaS) models, which allow low-skilled actors to launch large-scale attacks using readily available toolkits. In this context, the impact of such attacks is determined not only by the sophistication of the underlying technologies but also by the growing accessibility of digital crime ecosystems offering service-based attack platforms (Kumar & Upadhyay, 2023). The rising offensive capabilities suggest that defense systems lacking anticipatory design are increasingly vulnerable to adaptive and simultaneous attacks.

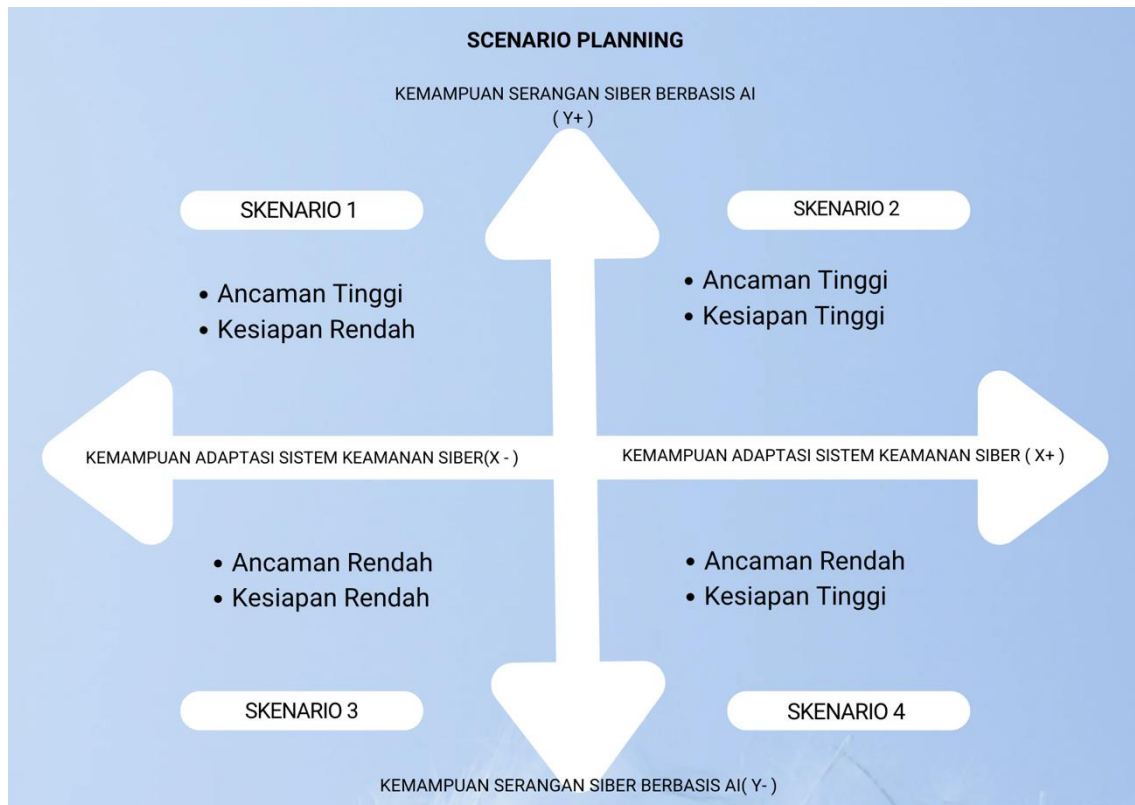


Figure 2.3. Scenario Planning of AI-Driven Ransomware Threats to Indonesian State-Owned Banks (2025–2029)

Source: Author's foresight analysis based on secondary data (2024)

Scenario 1: High AI-Driven Threats – Low Institutional Adaptability

This scenario represents the most critical condition for Indonesia's state-owned banking sector, wherein the intensity and technical complexity of AI-driven cyberattacks increase significantly, while institutional capacity to adapt cybersecurity defenses remains stagnant or even deteriorates. In such a context, threat actors are projected to rely heavily on AI-enhanced ransomware equipped with automated exploitation capabilities, lateral movement, simultaneous encryption of core systems, and evasion techniques designed to bypass traditional signature-based detection systems (Zhou, Lin, & Patel, 2023).

This vulnerability is compounded by internal factors such as incomplete implementation of cybersecurity audits, fragmented threat intelligence sharing between regulatory bodies and banks, and a lack of skilled personnel capable of detecting and responding to AI-based threats in real time. Under this scenario, attacks on core banking systems could disable essential services like ATMs, mobile banking, and national payment infrastructure, ultimately triggering a public trust crisis toward state financial institutions (Simorangkir, Purba, & Hartati, 2025).

Forecasts for the 2025–2029 period suggest a surge in the use of Ransomware-as-a-Service (RaaS) models, fueled by the low cost of launching attacks and high potential for financial gain. Given their role in managing public funds, distributing social

assistance, and maintaining the stability of national payment systems, Bank BUMNs are expected to remain prime strategic targets (Kumar & Upadhyay, 2023).

To mitigate this scenario, a transformational policy response is urgently required. This includes the integration of automated detection systems in alignment with ISO/IEC 27035 standards, the enhancement of Security Operations Center (SOC) functions, and the development of incident response protocols capable of initiating early containment within one hour of attack detection (International Organization for Standardization, 2018; National Institute of Standards and Technology, 2024). Furthermore, the policy roadmap must establish a 24-hour incident reporting deadline and accelerate the development of a national, real-time threat intelligence data center.

Failure to anticipate this trajectory through a foresight-driven approach could result in a systemic crisis within the national financial architecture. The consequences would extend beyond fiscal instability and disruption of public fund distribution; they would severely undermine long-term public trust in the integrity of the national banking system. Therefore, any policy intervention must transcend reactive measures and instead be based on anticipatory projections of evolving technologies and exponentially advancing threat patterns.

Scenario 2: High AI-Driven Threats – High Institutional Adaptability

This scenario illustrates a condition in which the frequency and complexity of AI-powered ransomware attacks continue to rise. However, state-owned banks (Bank BUMN) respond proactively with a high degree of cybersecurity readiness and integrated defense systems. Under this configuration, the banks have deployed advanced digital protection technologies such as automated threat detection, zero trust architecture, and AI-driven threat analytics capable of identifying anomalies before they reach critical infrastructure (Suvorova, 2023).

Although threat actors persist in employing sophisticated methods—including machine learning, fileless malware, and ransomware-as-a-service (RaaS) models that enable structured, simultaneous system penetration (CrowdStrike, 2025)—Bank BUMNs exhibit strong institutional readiness. This is evident through the reinforcement of Security Operations Centers (SOCs), continuous capacity-building of cybersecurity personnel, and the implementation of real-time incident reporting protocols aligned with international standards such as ISO/IEC 27035 (International Organization for Standardization, 2018).

Between 2025 and 2029, the velocity and automation of cyberattacks are expected to intensify, requiring strategic responses grounded in predictive analytics and pre-tested scenarios. Relevant mitigation policies under this scenario include establishing cross-institutional threat intelligence sharing networks among BUMNs and regulators, developing a national cybersecurity crisis command center for the banking sector, and conducting routine cyber resilience testing to evaluate emergency responsiveness to various forms of digital threats.

Despite the persistence of high-level threats, this scenario demonstrates that systemic risks can be significantly reduced when institutional defenses are both adaptive and responsive. Effective foresight strategies emphasize the importance of consistently strengthening institutional frameworks, ensuring transparent incident reporting, and integrating predictive data across sectors as a foundation for responsive and sustainable cybersecurity policymaking (Saritas, 2016; Linkov & Kott, 2019).

With sustained and continuous improvement in their defensive capabilities, state-owned banks have the potential to serve as a foundational pillar in strengthening the nation's cyber resilience architecture. This approach not only minimizes systemic disruption but also ensures service continuity and enhances public trust in national financial institutions amid the growing complexity of digital threats.

Scenario 3: Low AI-Powered Threats – Low Institutional Adaptability

This scenario describes a situation in which ransomware incidents targeting state-owned banks are minimal over a certain period. However, this decline is not accompanied by improvements in cybersecurity preparedness. The absence of large-scale attacks fosters a false sense of security, leading to diminished urgency in updating cyber defense architecture, conducting regular system vulnerability assessments, and enhancing human resource capabilities to detect latent and less visible cyber threats.

Under these circumstances, investment in cybersecurity systems tends to be neglected. Early detection mechanisms are underdeveloped, training for Security Operations Center (SOC) personnel is sporadic, and zero trust security frameworks are deprioritized. Incident reporting remains reactive and procedural, lacking integration with real-time systems. Additionally, incident response protocols are not aligned with international standards such as ISO/IEC 27035, which advocates structured and proactive incident handling processes (International Organization for Standardization, 2018). The absence of red team exercises, weak cross-agency threat intelligence sharing, and poor cybersecurity awareness at the executive level further amplify systemic vulnerabilities.

While attack frequencies may appear low, global trends indicate that cybercriminals often exploit such quiet periods to plan high-impact, precision-targeted attacks in the future (Suvorova, 2023). With advancements in AI and automation, these threats can be rapidly deployed at scale, making them harder to detect during initial stages (Zhou, Lin, & Patel, 2023). Without predictive, machine learning-enabled detection capabilities, institutional exposure to catastrophic breaches remains critically high.

Mitigation strategies for this scenario must begin with strengthening cybersecurity foundations. This includes developing baseline security controls, enhancing the quality and frequency of technical training, creating detailed incident contingency plans, and aligning procedures with global standards such as NIST SP 800-61 (National Institute of Standards and Technology, 2024). Regulatory reforms are also necessary to mandate regular security audits, thematic incident simulations, and enforce a 24-hour maximum reporting window for cyber incidents.

This scenario emphasizes that a lack of visible attacks is not a reliable indicator of digital system resilience. On the contrary, such stagnant periods may conceal structural weaknesses and latent threats that could trigger destructive consequences during future escalations. Therefore, applying a foresight-based approach remains essential—even in low-threat environments—to develop anticipatory strategies grounded in technological projections and evolving threat dynamics.

Scenario 4: Low AI-Powered Threats – High Institutional Adaptability

This scenario depicts an optimal condition in which ransomware threats targeting Indonesia's state-owned banking institutions (Bank BUMN) remain relatively low, while cybersecurity preparedness is consistently maintained at a high level. Rather than resulting in relaxed defenses, the low threat environment is leveraged as an opportunity to enhance system resilience through anticipatory measures and medium-term risk management strategies.

In this configuration, Bank BUMNs have implemented advanced digital defense systems grounded in zero trust architecture, machine learning-based anomaly detection, and adherence to international standards such as ISO/IEC 27001 and ISO/IEC 27035 for information security and incident response (International Organization for Standardization, 2018). Cybersecurity personnel undergo regular simulation-based training (e.g., red team exercises) and are continuously upskilled to stay ahead of evolving threat patterns. Real-time incident reporting protocols are connected to global response frameworks, including NIST SP 800-61 Rev. 3 (National Institute of Standards and Technology, 2024).

Despite the low threat landscape, institutions maintain predictive analytics and foresight scenario planning to anticipate potential future shifts in the attack surface. These proactive efforts include the institutionalization of inter-agency threat intelligence sharing, routine cyber resilience testing, and the establishment of cyber crisis command centers prepared to activate in response to escalations.

Sustained stability in this scenario is driven by alignment between adaptive regulatory policies and robust operational infrastructure. Key enabling factors include government incentives for institutional capacity building, standardized system interoperability, and risk-based cybersecurity approaches. Within this paradigm, cyber resilience is conceptualized not merely as incident response, but as a component of strategic learning embedded into organizational governance structures (Linkov & Kott, 2019).

During the projected 2025–2029 period, this scenario represents a model of best practices for managing cybersecurity under low-threat conditions. Maintaining consistent readiness allows Bank BUMNs to serve as a locomotive for strengthening the national digital resilience ecosystem. Their role becomes pivotal in ensuring the continuity of public services and fiscal stability when future threat escalations inevitably arise.

Roadmap

Within the foresight analysis framework, the current position of Indonesia’s state-owned banking sector (Bank BUMN) can be classified within **Quadrant I**—a condition characterized by high threat escalation but low adaptive capacity. According to the *2024 Indonesian Cybersecurity Landscape* report published by BSSN, ransomware variants such as LockBit, Ryuk, and various forms of Ransomware-as-a-Service (RaaS) have been actively targeting national strategic sectors. LockBit in particular emerged as the most dominant, with over 102,000 activities detected throughout 2024, reflecting the increasing use of artificial intelligence in attack distribution, penetration, and evasion mechanisms (Badan Siber dan Sandi Negara, 2024).

However, from the standpoint of cyber readiness within Bank BUMNs, institutional responses to these evolving threats have yet to demonstrate optimal adaptive capacity. Two key incidents underscore this structural vulnerability: the ransomware attack on Bank Syariah Indonesia (BSI) in May 2023, which disrupted banking services for several days (Chakravarti, 2023), and an unconfirmed but widely reported ransomware incident involving Bank Rakyat Indonesia (BRI) in late 2024 (Kompas, 2024). Indicators of systemic weakness include delays in incident reporting, the absence of internationally recognized incident response frameworks such as ISO/IEC 27035, and limited integration of threat intelligence sharing systems between banks and regulatory agencies (Simorangkir, Purba, & Hartati, 2025).

Given these dynamics, the strategic positioning of the Bank BUMN sector currently falls within a zone of elevated vulnerability. The high intensity of AI-driven cyber threats is not being met with commensurate levels of structural and tactical readiness to respond to, mitigate, or anticipate systemic attacks. This misalignment highlights the urgent need to formulate foresight-based prevention policies—ones that go beyond technical fixes to include institutional reform, regulatory alignment, and multi-stakeholder collaboration in building a national cybersecurity ecosystem.

Table 2.1. Cybersecurity Policy Roadmap for State-Owned Banking Institutions
Source: Author's foresight elaboration (2025)

Tahun	Fokus Strategis	Tindakan Utama	Target dan Indikator
2025	Deteksi Dini dan Standarisasi	- Adopsi ISO/IEC 27035 di seluruh bank BUMN	- Semua BUMN mengadopsi standar minimum keamanan informasi
		- Audit kesiapan cybersecurity tahunan berbasis NIST	- Pelaporan insiden <24 jam (OJK Compliance)
		- Pendirian tim respons insiden AI khusus ransomware	
2026	Kapasitas SDM dan	- Pelatihan SOC terhadap AI-based threat detection	- >90% personel SOC tersertifikasi

	<i>Infrastruktur SOC</i>	- Penambahan kapasitas SIEM (Security Information and Event Management)	- Sistem SIEM terpasang di 100% bank BUMN
2027	<i>Kolaborasi dan Threat Intelligence Sharing</i>	- Platform berbagi ancaman antar-BUMN dan regulator	- Threat sharing platform aktif dan real-time
		- Protokol koordinasi lintas sektor (Bank-BSSN-OJK)	- Simulasi bersama dilakukan dua kali setahun
2028	<i>Simulasi Skenario dan Ketahanan Resilien</i>	- Red team exercise berskala nasional	- Cyber Resilience Index sektor BUMN >80%
		- Simulasi serangan RaaS dengan penilaian preparedness	
2029	<i>Evaluasi dan Integrasi Strategi Nasional</i>	- Evaluasi roadmap dan integrasi ke Strategi Keamanan Siber Nasional	- Dokumen integrasi BUMN–nasional disahkan
		- Rencana kontinjensi siber berbasis skenario AI tingkat nasional	- Zero-day response plan diterapkan

Short-Term Roadmap (2025)

The initial phase in 2025 focuses on strengthening early detection systems and standardizing cybersecurity governance across all state-owned banking institutions (Bank BUMN). A strategic priority for this stage is the adoption of the international standard ISO/IEC 27035, which outlines systematic frameworks for detecting, responding to, and recovering from information security incidents efficiently (International Organization for Standardization, 2018). In parallel, annual cybersecurity readiness audits are introduced based on the National Institute of Standards and Technology (NIST) framework, aiming to assess compliance levels, operational preparedness, and the effectiveness of deployed IT security infrastructures (National Institute of Standards and Technology, 2024).

Moreover, 2025 marks a pivotal point for the establishment of a specialized Computer Security Incident Response Team (CSIRT) focusing on AI-powered ransomware. This team is expected to deliver rapid responses to incidents, with capabilities for early containment and real-time strategic escalation—particularly in addressing adaptive ransomware variants such as LockBit and Ryuk (Zhou, Lin, & Patel, 2023).

The key success indicators for this phase include full adoption of minimum cybersecurity standards across all Bank BUMNs and compliance with regulatory requirements set by OJK, particularly the 24-hour incident reporting deadline following initial detection. Through these measures, the technical and institutional foundations for mid-term cyber resilience are comprehensively established.

Roadmap 2026 – Capacity Building and Infrastructure Enhancement

The strategic focus in 2026 centers on strengthening both human resource capacity and supporting infrastructure to counter AI-driven cyber threats. A key priority is the implementation of intensive training programs for Security Operations Center (SOC)

personnel, specifically targeting competencies in machine learning-based anomaly detection and AI-assisted threat analytics. These training initiatives aim to sharpen technical responsiveness in identifying system anomalies and executing rapid mitigation. The target outcome for this year is for at least 90% of SOC personnel to hold nationally or internationally recognized certifications in AI-based threat detection (CrowdStrike, 2025).

Beyond human capital, institutions are also required to expand their Security Information and Event Management (SIEM) infrastructure to ensure accurate, large-scale, real-time anomaly data correlation. By the end of 2026, all Bank BUMNs are expected to have fully integrated SIEM platforms. These systems play a pivotal role in enabling automated threat detection and in generating predictive risk dashboards (Suvorova, 2023).

By reinforcing both the human and technological pillars of cybersecurity, institutions will be well-positioned to enter the next phase of the roadmap, which emphasizes interagency collaboration and system interoperability.

Roadmap 2027 – Building a Collaborative Threat Intelligence Ecosystem

In 2027, the national cybersecurity strategy shifts toward reinforcing a collaborative ecosystem, with a specific focus on developing real-time threat intelligence sharing frameworks. State-owned banks (Bank BUMN) and financial regulators are mandated to establish interoperable platforms that enable continuous exchange of technical data, including attack metadata, adversary tactics, and indicators of compromise (IOCs). As highlighted by CONCORDIA (2021), the effectiveness of large-scale incident responses is highly contingent on the timeliness and integration of cross-institutional threat intelligence.

Furthermore, formalized inter-sectoral coordination protocols are introduced between Bank BUMNs, the National Cyber and Crypto Agency (BSSN), and the Financial Services Authority (OJK). These protocols define mechanisms for incident escalation, task distribution during cyber emergencies, and contingency communication procedures.

Key performance indicators for this year include the successful execution of biannual joint simulation exercises, involving all primary stakeholders in Indonesia's financial sector. These simulations aim to enhance institutional readiness, validate emergency response coordination, and accelerate response times during potential systemic cyberattacks.

Roadmap 2028 – National Cyber Resilience Simulation and Proactive Readiness

In 2028, the strategic focus transitions toward large-scale cyber resilience testing through scenario-based simulations at the national level. This involves red team exercises designed to evaluate institutional responsiveness under simulated Ransomware-as-a-Service (RaaS) attack conditions. These simulations aim to test the robustness of defense systems, assess the efficiency of internal and external coordination, and examine institutional readiness against latent, automated cyber threats (Zhou, Lin, & Patel, 2023).

A Cyber Resilience Index becomes the primary performance metric for this phase, with a targeted minimum score of 80% across the Bank BUMN sector. The results from

these simulations serve as empirical input for refining security protocols and issuing adaptive policy recommendations based on evolving adversarial tactics.

This year marks a critical milestone in shifting from a reactive security posture to a scenario-based proactive strategy, supported by structured and documented preparedness plans (Saritas, 2016). The transition emphasizes measurable strategic foresight and institutional agility in anticipating future threat landscapes.

Roadmap 2029 – Strategic Integration and National AI Contingency Planning

The final phase of the roadmap in 2029 is centered on comprehensive evaluation and the full integration of previous initiatives into Indonesia's national cybersecurity policy framework. This stage involves assessing all milestones, challenges, and dynamics of the roadmap's implementation from 2025 onward. The outcomes will be aligned and embedded into the broader National Cybersecurity Strategy, forming a formal government document to guide long-term efforts in safeguarding the national financial sector (Linkov & Kott, 2019).

In addition to the macro-level evaluation, this year also emphasizes the development of a national AI-based scenario contingency plan. The plan includes protocols for zero-day response, systemic cyber recovery mechanisms, and the establishment of an AI Threat Response Coordination Center. The implementation of a real-time zero-response protocol is positioned as the primary success indicator, representing Indonesia's full readiness to counter escalating AI-driven ransomware threats through a unified, sustainable, and forward-looking national strategy (National Institute of Standards and Technology, 2024).

Conclusion

AI-enhanced ransomware has emerged as a strategic cyber threat with increasing sophistication against the digital infrastructure of Indonesia's state-owned banking sector. Unlike earlier attacks which were largely static and structured, recent ransomware

variants such as LockBit, Ryuk, and Blocker have evolved into adaptive, automated threats capable of bypassing traditional signature-based defenses and endpoint detection systems. According to BSSN's 2024 annual report, more than 102,000 LockBit-related activities were detected, with the financial sector identified as a primary target (Badan Siber dan Sandi Negara, 2024), highlighting the systemic vulnerability of Bank BUMNs to these evolving threats.

Despite the severity of the threat landscape, the current cybersecurity readiness of Bank BUMNs remains inadequate. The five-day service outage caused by the LockBit 3.0 attack on Bank Syariah Indonesia in 2023, alongside a similar incident reported at Bank Rakyat Indonesia in 2024, revealed critical weaknesses in incident reporting protocols, institutional response readiness, and the absence of real-time incident handling mechanisms. Although regulations such as POJK No. 11/2022 and Presidential Regulation No. 82/2022 are in place, they have yet to comprehensively address AI-based attacks or establish a cross-institutional threat intelligence sharing ecosystem (Simorangkir, Purba, & Hartati, 2025; OJK, 2022).

Horizon scanning conducted in this study identified an annual trend of heightened ransomware activity during the final fiscal quarter, particularly in November and December 2024, during which BSSN recorded over 160 million cyber anomalies. This surge correlates with increased digital transactions and decreased institutional vigilance during national holidays. Thus, the need for predictive early warning systems and integrated institutional preparedness—linking policy with technical protocols—is increasingly urgent.

Three dominant threat drivers were identified: the rising automation of AI-driven ransomware, fragmentation in cybersecurity regulatory frameworks, and the growing accessibility of Ransomware-as-a-Service (RaaS) on the dark web. RaaS models lower the technical barrier for malicious actors, broadening the attacker spectrum and amplifying the scope and intensity of cyberattacks (Kumar & Upadhyay, 2023). These dynamics underscore the urgency for foresight-driven policy responses that are both anticipatory and tailored to evolving threat patterns.

Scenario mapping places Bank BUMNs within Quadrant I—a condition characterized by high threat intensity and low adaptive capacity. This position denotes maximum institutional vulnerability to AI-driven ransomware. To address this, the study

proposes a five-year policy roadmap (2025–2029) designed to shift Bank BUMNs toward Quadrant II, emphasizing technical transformation, institutional reinforcement, and scenario-based policy integration (Saritas, 2016; Linkov & Kott, 2019).

The roadmap includes five progressive stages: adoption of ISO/IEC 27035 and establishment of an AI-focused CSIRT in 2025; SOC capacity building and SIEM integration in 2026; threat intelligence sharing and joint simulation exercises in 2027; national-level cyber resilience testing in 2028; and the consolidation of crisis command infrastructure and policy synchronization in 2029. Each stage is aligned with performance indicators and threat escalation data to guide implementation.

The application of foresight-based strategic intelligence in this research plays a pivotal role in shifting the cybersecurity paradigm from reactive to proactive. The integration of horizon scanning, trend analysis, and scenario planning equips financial institutions with more precise and risk-informed policy tools (Prunckun, 2010; Saritas, 2016).

Ultimately, the study's findings extend beyond institutional concerns to influence the design of national security policy. As custodians of public funds and operators of national payment systems, the cyber resilience of Bank BUMNs directly impacts fiscal stability and public trust. Building a cybersecurity defense that is technologically robust, institutionally responsive, and regulatory adaptive is imperative for addressing the challenges posed by rapidly evolving AI-based digital threats.

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